

AI-Guided Exploration of Arithmetic Dynamical Systems:

Constraints, Failure Modes, and Search Strategies toward Hilbert–Pólya Structures

Liang Wang

School of Artificial Intelligence and Automation, Huazhong University of Science and Technology

Wuhan 430070, P.R. China

wangliang.f@gmail.com

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Conference-oriented methodology paper; source-bound Phase-I case study

Abstract

Recent advances in AI-assisted mathematical discovery, program search, and formal reasoning motivate a practical design question: how can a long-horizon open problem be reorganized into a broad, falsifiable, and evidence-accountable human–AI search? We present a prospective candidate-engineering framework, grounded in a source-bound retrospective Phase-I design case on arithmetic dynamical structures relevant to the Hilbert–Pólya programme. The paper does not claim a proof of the Riemann Hypothesis, a completed Hilbert–Pólya realization, or a spectral identification of Riemann zeros.

The framework has three layers. In Layer 1, a mathematician fixes a mathematical origin, permitted data, and an admissible transformation class; the present case uses a prime-symbolic origin, whereas a new mathematician-supplied basic idea begins a new Layer-1 record. In Layer 2, mathematician and AI jointly draft and version-freeze an AGENTS.md contract specifying candidate identity, lineage, gates, controls, portfolio policy, and stopping rules. In Layer 3, bounded AI work returns a source-linked evidence handoff for mathematician review. The accompanying roadmap requires a single candidate to establish an endogenous arithmetic source, orbit/repetition ownership, and a same-object zeta or determinant before stronger analytic and operator obligations are considered.

The case comprises one early exploration entry and five system-family sessions. Its P1 Wiki retains catalogued provenance records rather than a common score of completed papers, theorems, or Route advances. No source-bound top-level direction documents an integrated same-object $A_0+A_1+A_2$ chain in one natural candidate. We therefore motivate a broader but lineage-filtered candidate portfolio, screened early for arithmetic, clock, and determinant ownership rather than assembled retrospectively from partial successes. The associated Wiki is a provenance and navigation resource designed to preserve positive local results, controls, scoped failures, and reuse boundaries for subsequent researchers.

Keywords: arithmetic dynamics; human–AI collaboration; Hilbert–Pólya programme; periodic orbits; dynamical determinants; research methodology; reproducibility

Scope and nonclaims. This manuscript is a conference-oriented methodology paper supported by a source-bound research record. It reports a framework, local results, controls, and open obligations. It does *not* establish the Riemann Hypothesis, a prime-power trace formula, a completed- Ξ determinant identity, a self-adjoint Hilbert–Pólya operator, or a route verdict transferred across distinct objects.

1 Introduction

The Riemann Hypothesis (RH) remains an unusually demanding target for human–AI mathematical exploration. As of 14 September 2026, the Clay Mathematics Institute continues to classify RH as an unsolved Millennium Prize Problem [3]. The challenge is not simply to generate a numerical resemblance or an appealing operator. In the candidate class and evidence-obligation architecture adopted here, a credible

Hilbert–Pólya (HP)-oriented candidate is required to connect arithmetic information, periodic dynamics, analytic structure, and an operator-theoretic trace identity without changing the mathematical object along the way.

This difficulty is timely in light of recent, but carefully bounded, advances in AI-assisted mathematics. Machine learning has helped human researchers formulate and investigate conjectures in selected areas of mathematics [5]; evaluator-coupled language-model search has generated new constructions in constrained combinatorial and algorithmic settings [7]; neuro-symbolic systems have solved 25 of 30 olympiad-level Euclidean-geometry problems [10]; and reinforcement learning has been used to search for Lean-verified proofs in a specialized olympiad-level setting [6]. Most directly relevant to RH, a 2026 preprint reports an unconditional critical-line proportion bound for zeta zeros together with Lean 4 formal verification [1]. These examples show that AI can contribute to workflows with clear representations, evaluators, and verification boundaries. They do *not* constitute a proof of RH, a solution of any other active Millennium Prize Problem, or evidence that a language model can replace mathematical validation.

These strands supply important components—mathematical conjecture support, evaluator-coupled search, formal proof checking, and reproducibility principles [5–7, 9, 10, 12]—but they do not by themselves prescribe how a long-horizon arithmetic-dynamical candidate should inherit a mathematical origin, retain one object identity across orbit and determinant constructions, or preserve a scoped failure for later reuse. The present paper addresses that narrower design problem rather than proposing a general account of mathematical intelligence.

The question addressed here is therefore methodological:

How can the search for an arithmetic dynamical structure relevant to the HP programme be reformulated as a broad, falsifiable, auditable, and cumulatively reusable human–AI candidate-search problem?

The answer developed from the Phase-I record is *candidate engineering*. Rather than optimize a loosely specified analogy to the Riemann zeros, the researcher freezes a candidate’s identity and tests an ordered set of obligations: endogenous arithmetic source, primitive-orbit and repetition ownership, a dynamical zeta or determinant belonging to that same object, and only then the stronger analytic and operator-theoretic requirements. AI can help enumerate, compare, document, and challenge candidates; human researchers remain responsible for definitions, proof acceptance, claim authorization, and external communication.

This paper makes six contributions.

1. It puts a three-layer, human-governed framework first: the mathematician owns the research origin and scientific authority; a jointly drafted, human-approved AGENTS.md makes the search contract explicit; and AI returns a bounded evidence packet for review.
2. It places the updated programme roadmap beside that framework as an *obligation architecture*, not a cumulative progress chart or a workflow score.
3. It makes the prime-symbolic origin concrete through a two-lane intellectual genealogy: five internal prior-work threads constrain the admissible transformation class, while a prime-side operator result supplies an external acceptance benchmark rather than inherited credit.
4. It inventories the heterogeneous Phase-I portfolio as one free entry point and five system-family sessions, separating catalogue scale, bounded focal records, reusable local outputs, and unresolved bridges.
5. It formulates the *same-object* invariant and a candidate card that make hidden ownership changes, post-hoc prime attachment, and premature operator claims visible early.
6. It treats the versioned Wiki as a provenance and navigation resource designed to preserve successful local work, controls, scoped failures, and reuse boundaries for future candidate engineering.

The roles of these contributions are distinct. The primary contribution is a prospective candidate-engineering framework; the Phase-I material is a source-bound retrospective design case from which its constraints are distilled; and the Wiki is a navigation-and-provenance resource designed to support later assessment and possible reuse. The paper does not claim that Phase I was a controlled evaluation of the framework or that its six directions followed one historical AI protocol.

The detailed summary of each direction is placed in appendix A; the repository remains the canonical navigation layer for the underlying source materials [11]. The present manuscript is a source-bound methodological case study, supported by a research record rather than an independent re-proof of those local results.

2 A human-governed framework for AI-assisted arithmetic-dynamical search

Figure 1 is the paper’s core methodological contribution: a human-governed framework for bounded AI search, organized in three layers. *Layer 1* gives the mathematician responsibility for the prime-symbolic research origin, the scientific question, permitted arithmetic data, and the admissible search envelope. *Layer 2* turns those choices into a jointly drafted, human-approved AGENTS.md contract. *Layer 3* requires bounded AI execution, an explicit evidence handoff, and mathematician review before another round can be authorized.

The prime-symbolic origin shown in figure 1 is the present case-study instantiation of Layer 1, not a permanent restriction on the framework. If a mathematician introduces a different basic mathematical idea, that change creates a new Layer-1 origin record and requires a new or revised, human-approved and version-frozen Layer-2 contract before AI-assisted search can begin. It is not an undocumented broadening of an existing candidate portfolio.

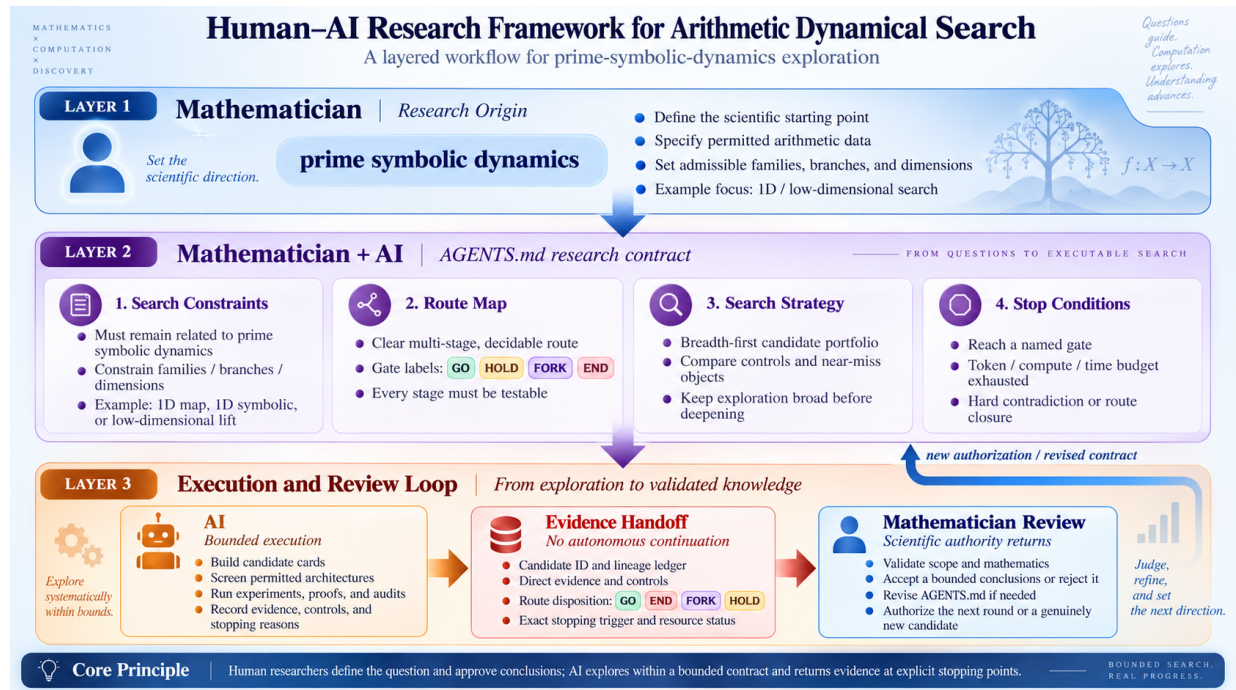


Figure 1: Three-layer human-AI research framework for arithmetic-dynamical search. *Layer 1*: the mathematician fixes the prime-symbolic origin and permissible search envelope. *Layer 2*: a jointly drafted, human-approved AGENTS.md fixes constraints, a testable route, a breadth-first portfolio with controls and near misses, and stopping conditions. *Layer 3*: bounded AI execution, evidence handoff, and mathematician review; continuation needs a new authorization or revised contract. GO, HOLD, FORK, and END are unordered workflow dispositions, not Route-A/Route-B passes. This is a prospective Round-2 framework, not a claim that Phase-I sessions used one uniform agent protocol.

This structure has a restrictive admission rule: no generic chaotic or quantum system may be admitted and decorated with primes later; nor may AI enlarge the admissible class, replace a missing preservation mechanism, or authorize a mathematical claim.

The Layer-2 contract has four irreducible parts; Layer 3 operationalizes them through bounded execution,

evidence handoff, and mathematician review. First, the search-constraints card fixes the admissible class. For this programme, a main candidate must trace to a prime-symbolic ancestor and name a preservation, deformation, dimensional lift, or geometric-realization mechanism. It must also state the permitted system families, branches, dimensions, parameter sources, and arithmetic inputs before the search starts. The illustrative one-dimensional or low-dimensional focus in figure 1 is a declared family restriction, not a license to attach prime language to any available dynamical system. Any later non-autonomous deformation or conservative lift must preserve, transform, or explicitly refute a named piece of that lineage.

Second, the route-map card contains a multi-stage route that can be decided at its stated scope. It specifies the evidence required for a gate, the controls that can falsify a proposed mechanism, and a non-ordered set of possible workflow dispositions: GO permits the next declared screen; HOLD returns an unresolved scientific decision to the mathematician; FORK creates a new object identity and candidate card; and END archives a bounded result. Every stage must be testable. These labels cannot be used as substitutes for A0–A4 or Route-B evidence.

Third, the search-strategy card makes breadth first the default. AI can compare several permitted architectures, including controls and near-miss objects, and keep source and ownership ledgers synchronized before deeper work begins. That breadth is disciplined: each architecture must pass the lineage admission test and cheap same-object checks. The strategy is not a claim that breadth is universally optimal; it is a practical response to the Phase-I finding that local successes from distinct objects do not add to one candidate.

Fourth, the stop-conditions card makes operational boundaries explicit. A search may stop at a named roadmap gate, a hard contradiction or declared route closure, a loss of lineage or same-object ownership, a scoped negative result, or exhaustion of its specified token, compute, or time budget. Budget exhaustion is an operational stopping condition, not a mathematical negative result. In Layer 3, AI may perform only contract-permitted candidate screening, experiments, proof attempts, and audits. At a stop it hands the mathematician a compact evidence packet containing the candidate ID, lineage ledger, direct evidence and controls, route disposition, exact stopping trigger, and resource status. The mathematician then validates scope and mathematics, accepts or rejects the bounded conclusion, and may issue a new authorization or a revised contract for a further round. There is no autonomous continuation. A versioned AGENTS.md therefore supports traceable, reproducibility-oriented research governance; it is not mathematical evidence and cannot validate a proof by itself.

The framework is prospective. The prior-work guide records a research trajectory and explicitly does not claim to have been a historical candidate selection manual. The framework uses that trajectory to govern future candidate engineering without retroactively assigning a uniform AI workflow to the six Phase-I directions [11].

2.1 The companion mathematical obligation roadmap

Figure 1 answers who fixes the search space, who executes bounded work, and when scientific authority returns to the mathematician. The roadmap in figure 2 answers a different question: which mathematical obligations an admitted candidate must earn. Neither diagram substitutes for the other.

Route A begins with A0, an endogenous arithmetic source. A1 requires primitive, reproducible orbit ownership and a stated repetition convention. A2 asks for a stable dynamical zeta or Fredholm determinant whose operator, domain, normalization, and orbit expansion belong to that same candidate. A3 and A4 add the relevant global analytic/Weil-compatible structure and a natural classical-to-quantum lift. Route B is stricter still: it concerns a fixed operator and domain (B1), self-adjointness (B2), the target spectral type (B3), an exact prime-power/Weil-compatible trace identity (B4), and a global completed- Ξ determinant or divisor equality (B5). It cannot repair a weak Route-A object.

The roadmap's key word is *same*. Under the *same-object* screening criterion adopted here, a symbolic clock, a physical closed orbit, a determinant borrowed from a different transfer system, and a separately constructed operator do not count as components of one candidate unless common ownership, or a precisely

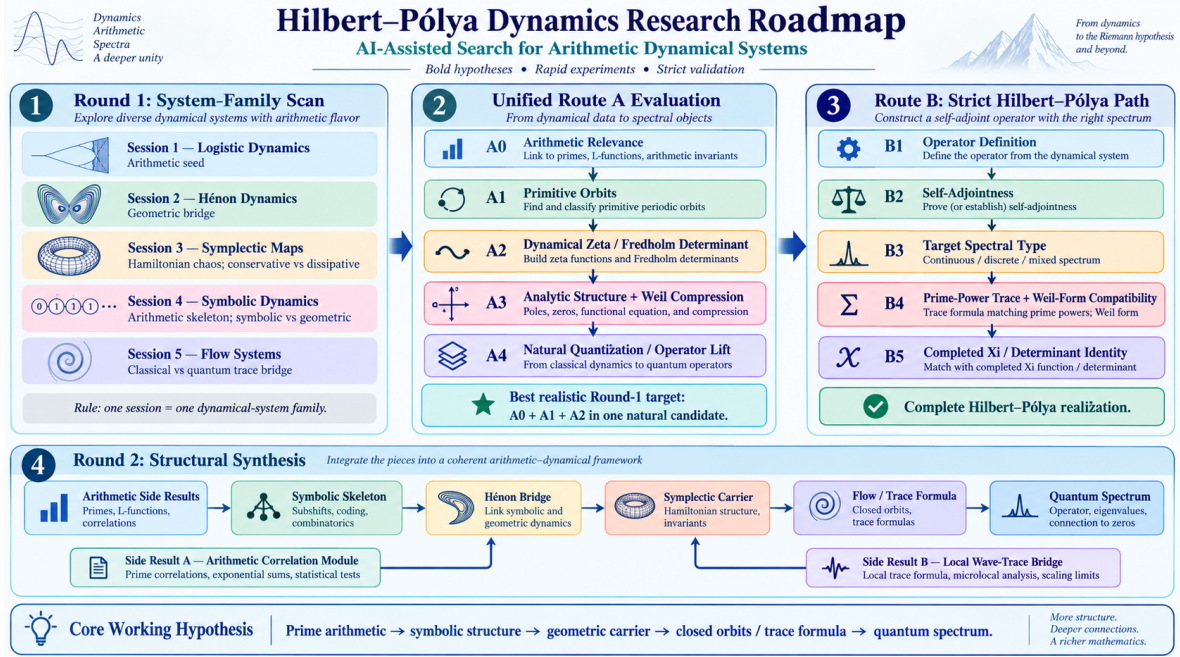


Figure 2: Author-proposed search and evidence-obligation roadmap. The roadmap separates a system-family scan, Route-A candidate obligations, the stricter Route-B operator path, and a Round-2 structural-synthesis hypothesis. Arrows denote evidence that must be established *within one candidate*; they do not report completed gates, transfer credit across systems, or establish an HP realization or RH. The Round-2 panel is a lineage-constrained new-object design hypothesis, not a post-hoc assembly of Phase-I outputs; every link requires a preservation ledger and new *same-object* validation. The raster asset is a byte-preserved, release-local mirror of the author-supplied roadmap (SHA-256 f5e70c5120474702e4f0715bce90c7a88fd5936164594e8b37a2ee0993549d1d).

established identification or intertwining relation, is supplied:

Same-object invariant
Arithmetic source, clock/repetition, orbits, determinant, and operator must have a common owner or an explicitly established identification.

Thus GO, HOLD, FORK, and END are workflow dispositions under the human-approved contract; A0–A4 and B1–B5 are candidate-specific mathematical obligations. The remainder of the paper instantiates this distinction through the three layers of figure 1.

3 From Hilbert-Pólya intuition to an obligation architecture

The HP programme is often motivated by the expectation that a suitable spectral or trace structure could illuminate the nontrivial zeros of the zeta function. That motivation has inspired important connections among spectral asymptotics, trace formulae, and arithmetic [2, 4]. Dynamical zeta functions also have a well-established role in the study of expanding maps and Anosov flows [8]. None of these conceptual links licenses a candidate-specific conclusion by itself. The central practical question is how one turns that broad intuition into an origin-constrained, contract-governed, and source-bound research programme. The following subsections follow the three layers of figure 1: mathematical origin, executable research contract, and historical evidence/handoff.

3.1 Layer 1: mathematical origin and an admissible transformation class

Layer 1 is the non-delegable mathematical choice that makes the search space meaningful. In this case study, the starting point is not an arbitrary chaotic or quantum model; it is the project’s earlier work on

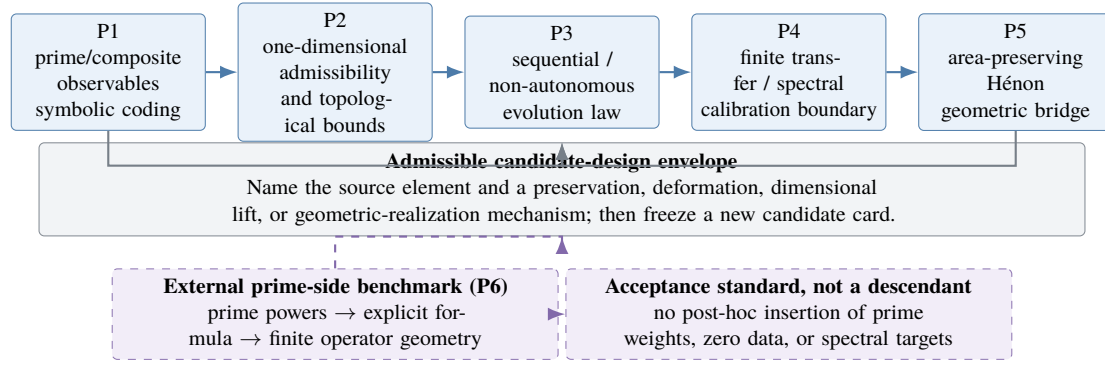


Figure 3: Layer-1 research origin and candidate-design boundary. P1–P5 are an internal conceptual genealogy from the prior-work guide; P6 is a distinct prime-side benchmark. The arrows express research provenance and admission constraints, not theorem implications, Route transfer, or evidence that a proposed descendant already has a prime-dynamical realization.

prime/composite observables, symbolic coding, and the structural limits of low-dimensional dynamics. The first five prior-work papers form an internal conceptual genealogy. The sixth paper is intentionally shown separately: it is a prime-side arithmetic/operator benchmark, not a mathematical descendant of the first five [11].

Table 1: Layer-1 prior-work lineage ledger. P1–P5 are project prior-work source records summarized by the guide; this paper does not independently re-review their mathematical claims or publication status. P6 is the separately cited external preprint benchmark. The final column states an admission constraint, not a theorem transfer.

Prior source	Retained mathematical idea	Admissible next change or boundary
P1: <i>The Emergence of Prime Distribution from Low-Dimensional Deterministic Chaos</i>	Prime/composite observables and symbolic coding supply the arithmetic-dynamical seed.	Preserve a named observable or coding; a numerical resemblance alone does not supply a spectral identity.
P2: <i>Transient Chaos and Topological Bounds in Prime Dynamics</i>	Finite-word, admissibility, and one-dimensional topological obstructions turn failed constructions into information.	Use the obstruction as a rejection test or justify a lift; do not assume that a one-dimensional encoding is complete.
P3: <i>A Sequential Birkhoff Theorem</i>	Sequential/non-autonomous evolution is a mathematical object with its own statistical conditions.	Admit a time-varying deformation only with an explicit evolution law and a stated preserved quantity.
P4: <i>Non-Autonomous Logistic Spectral Experiments</i>	Finite transfer/spectral calculations distinguish calibration from intrinsic chronological-orbit structure.	Use finite spectra as controls or diagnostics; they cannot replace a same-object determinant or target identity.
P5: <i>An Area-Preserving Hénon-Map Model</i>	Area-preserving Hénon geometry motivates a conservative dimensional bridge.	A geometric lift must name its projection, return relation, invariant, or open preservation ledger.
P6: <i>More Than Two Thirds of the Zeta Zeros Are Simple and on the Critical Line</i> [1]	Prime powers, the explicit formula, and finite operator geometry meet in a separate external arXiv preprint used here as a scoped arithmetic-side benchmark.	External benchmark, not a P1–P5 descendant; it motivates the paper’s no-post-hoc-insertion admission rule.

For any candidate, the card must identify its ancestral node or nodes, the transformation class being used, and the exact mechanism that is preserved, deformed, lifted, geometrically realized, or refuted. The mechanism may be a specified prime-symbolic observable, admissibility rule, return structure, coding, clock, or rigorously defined deformation thereof. If the relation is unproved, it is recorded as OPEN; it cannot be replaced by a narrative similarity. Thus a generic area-preserving map, fashionable arithmetic space, trace formula, or operator becomes an external control unless it carries this ledger. A non-autonomous model must likewise state an evolution law rather than merely fit a time-dependent parameter.

The P6 benchmark has a different role. It keeps prime powers and the explicit-formula side visible as a

scoped acceptance reference, but it supplies no transfer of prime tables, von Mangoldt weights, or zero data into a distinct dynamical object. This distinction is designed to make the common reversal of choosing a generic system first and attaching arithmetic language only after the fact visible at candidate admission.

3.2 Layer 2: a versioned research contract

Layer 2 makes Layer 1 executable without allowing the executable plan to silently redefine the mathematics. A jointly drafted, human-approved `AGENTS.md`, paired with a plan and candidate cards, records four contract cards before bounded work begins. It is a research-governance artifact: it specifies what the AI may screen and report, but never certifies a mathematical conclusion.

For a round to be replayable as a governance record, the operative contract must be fixed by a repository commit or tag and linked to its `plan.md`, candidate card, declared inputs, and evidence ledger. A mutable working `AGENTS.md` is useful navigation and planning material, but it is not a frozen evidence package by itself.

Table 2: Concrete Layer-2 contract cards. An operative contract is versioned before execution; the table states a prospective protocol and does not claim that all Phase-I materials historically used one uniform `AGENTS.md`.

Contract card	Record fixed before search	Human authority retained
Search constraints	Basic mathematical idea; permitted arithmetic data; lineage anchor; allowed system families, branches, dimensions, and transformations; required preservation/deformation/lift/geometric-realization ledger.	Choose or revise the research origin and decide whether a candidate is main portfolio material or an external control.
Testable route	Candidate identity; named evidence gates; controls; GO, HOLD, FORK, and END dispositions; required same-object checks.	Approve gate meanings and decide whether the evidence actually earns a disposition or a mathematical claim.
Portfolio strategy	Breadth-first ordering among lineage-admissible architectures; designated positive/negative controls; a rule for when a new identity is a fork rather than an extension.	Set search priorities, resource allocation, and the scientific value of a near miss or a scoped failure.
Stop and handoff	Gate-specific stopping triggers; contradiction, loss of lineage, ownership failure, route closure, or declared token/compute/time budget; handoff fields and release status.	Validate the stopping scope, accept/reject the report, and authorize any new round, public claim, or release.

For this project, the contract requires that each candidate state one owner for phase space, map or flow, roof or clock, primitive-orbit/repetition convention, arithmetic mechanism, coding, transfer/determinant object and domain, controls, and any later quantum operator. A loss of that identity is a FORK or END, not an invitation to borrow a missing component. The contract also makes the default portfolio breadth-first after the lineage screen; this is a strategy proposal motivated by the case study, not a retrospective claim that Phase I was a controlled breadth-first experiment.

The practical value of AI in this setting is not an ungrounded “guess the answer” mode. It is most credible where the representation and rejection condition are exposed: candidate enumeration, source retrieval, object-ledger checking, adversarial control generation, proof-dependency review, and reproducible documentation. This perspective is close to the generator–evaluator separation demonstrated in program-search work [7], but the evaluator for a mathematical research candidate is partly formal and partly human. It includes source identity, mathematical scope, and whether a purported bridge preserves the object.

Human authority remains indispensable. A fluent synthesis can make nonidentical constructions look continuous; an automated retrieval system can surface a nearby theorem under a misleading normalization; a numerical fit can conceal target leakage. The human researcher must decide whether definitions are

genuinely the same, whether a proof is valid, which claims have been earned, and whether a record may be released.

3.3 Layer 3: bounded execution, evidence handoff, and source-bound record

3.3.1 Case-study and evidence-synthesis protocol

This paper is a source-bound case-study synthesis, not a systematic review, a benchmark of research strategies, or an experimental comparison of AI systems. Its unit of analysis is a top-level research direction, rather than an individual file, PDF, build, or package identity. The case set is fixed by the Phase-I repository organization at the historical evidence snapshot: one early free-exploration entry and five subsequently organized system-family sessions. It is therefore a complete six-direction portfolio at that stated scope, not a random or statistically representative sample of arithmetic dynamics.

The synthesis uses three deliberately non-interchangeable information layers. First, direction-level mathematical statements are bounded to the historical source-corpus snapshot 419ee36c1e3. . . . Second, navigation and catalogue-size statements use the later P1 Wiki conversion snapshot afe747a194b9. . . . The full identifiers are recorded in figure 4 and appendix D. Third, external references provide motivation and context only. For every portfolio row, the paper records the focal object or batch, a bounded reusable output, and a named discriminating boundary, then links the reader to the controlling direction pages in appendix A. File counts are never used to infer theorem strength, novelty, resource use, or Route progress.

The principal coding rule is non-compensation. A direction is described as having an integrated $A0+A1+A2$ chain only if its own source-bound record documents all three obligations for one natural candidate with one object identity. A local result, an internal process state, an external archive, or a result in another direction cannot supply a missing obligation. Conversely, the absence of such a documented chain in this synthesis is not an impossibility claim about the whole family. This rule permits negative or inapplicable outcomes to remain useful without turning them into a common score.

3.3.2 The Phase-I corpus and its status vocabulary

The Phase-I knowledge base separates one early free-exploration line, `zeta_mvp0`, from five roadmap-driven sessions: Logistic Dynamics, Hénon Dynamics, Symplectic Map, Symbolic Dynamics, and Flow Systems [11]. The five P24–P28 continuous-time forms belong to the Flow session; they are not an additional top-level direction. The detailed historical synthesis is bounded to the stated source-corpus snapshot 419ee36c1e310469209f7b83c096ec8aea448386 (13 September 2026 UTC). The new paper and the updated roadmap are later documentation artifacts and do not retroactively alter that historical record.

Four kinds of statements are kept distinct throughout this paper:

Kind	Meaning and non-equivalence
Source identity	Where a file, claim, source lock, or imported archive comes from; it is not independent mathematical verification.
Process / record	A build, receipt, frozen contract, or completed batch; it is not automatically a theorem, Route pass, or publication.
Local mathematics	A result for a named object and scope; it is not a global zeta/zero/HP conclusion.
Route assessment	A gate evaluation for a typed candidate; it cannot transfer to another object or session.

This separation is especially important for AI-assisted work. It is designed to guard against a model treating a polished PDF, a large corpus, or a familiar keyword as an authorized conclusion. It also makes it possible to retain the value of a local theorem or a scoped negative control when it does *not* close the full route.

3.3.3 Evidence handoff, contract roles, and historical boundary

Historical and prospective records must remain distinguishable. The original research lines were created under their own mathematical and project protocols; this paper does not rewrite history by claiming that they all used one uniform AI procedure. For the prospective framework in figure 1, the following table makes the division of labor operational. AI can prepare and check a bounded search, while the mathematician controls its scientific scope, mathematical validity, and authorization to continue or release.

Table 3: Operational division of labor under the prospective framework in figure 1. The AGENTS.md contract constrains AI action but does not replace mathematical validation.

Stage	AI-assistible work	Human responsibility
Origin and constraints	map prior work; surface permissible branches; draft candidate-card and protocol options	choose the prime-symbolic origin; define the scientific question, permitted data, families, branches, and dimensions
AGENTS.md contract	formalize constraints; check decision labels and stopping clauses for consistency	approve and version-freeze the operative contract, its gate meanings, and its authorization boundary
Bounded search	enumerate eligible architectures breadth first; map sources, owners, and controls; prepare evidence ledgers	validate definitions, proofs, computations, preservation claims, and any GO decision
Stop, handoff, and release	record exact triggers; prepare the decision packet, link/citation checks, and reproducible artifacts	accept or revise a bounded conclusion; authorize a fork, renewed work, claims, authorship, and release

The record is intended to be inspectable rather than merely persuasive. The repository preserves the navigation pages, original project materials, conclusion boundaries, and accompanying research-record paper [11]. A clean build or an internally complete batch is therefore reported only as a process fact. It does not substitute for a proof check, external peer review, or a Route verdict.

3.3.4 Layer 3 portfolio: six-track source-bound evidence

The six lines produced different kinds of useful output. They cannot be ranked as six positions along one common route, because their objects, evaluation schemas, and scope boundaries differ. Figure 4 therefore uses colors to distinguish evidence roles, not to imply a success score: blue marks a preserved local capability, purple a local analytic or operator-related structure, gray a route-neutral or non-comparable state, and red a named missing bridge or stop boundary.

P1 Wiki inventory snapshot: 1,126 logical package identities across one early free entry and five system-family sessions. Counts identify reader/navigation records, not completed papers, theorems, resource expenditure, Route credit, or distance to RH.			
Track	Inventory and focal record	Bounded reusable output	Key boundary
zeta_mvp0 early free entry	405 logical records native work plus source-preserving imported archive	local counting; fixed self-adjoint compact-resolvent convention; relative trace	endogenous prime-time and von Mangoldt $\tau \log p$ bridge remains open
Session 1 Logistic	31 logical records focal fixed-parameter analytic chain	same-object return–roof–transfer–Fredholm chain	arithmetic origin, Riemann-target closure, and quantization absent; root hunting stopped
Session 2 Hénon	425 logical records focal C424–C428 sealed; C429–C433 admitted	restricted periodic classifications and low-/higher-dimensional geometric diagnostics	A0 and A1 weak or scoped; no documented same-object A2–A4 closure
Session 3 Symplectic	31 logical records five focal local interfaces	conservative-geometry interfaces and diagnostic tools	Route status is NOT_APPLICABLE; freeze a candidate before gate claims
Session 4 Symbolic	199 logical records P211–P215 exact-five internal complete	exact finite-system results and an audit-friendly evidence pattern	no natural geometric carrier, global analytic owner, or operator closure
Session 5 Flow	35 logical records focal P24–P28 five continuous-time forms	clock, roof, trace, and ownership stress tests	for P24–P28 only: positive arithmetic $A_2 = 0/5$; Route-B invocation = $0/5$
Portfolio-level boundary: no source-bound top-level direction documents an integrated same-object $A_0+A_1+A_2$ candidate. Local results from distinct rows are non-additive.			

Figure 4: Phase-I portfolio overview: inventory, focal records, bounded outputs, and discriminating boundaries.

The upper band separates the early `zeta_mvp0` entry from the five subsequently organized system-family sessions. The inventory comes from the P1 Wiki conversion snapshot `afe747a194b92b9804bd69e2f57696884e47d53b`; the historical mathematical synthesis is separately bounded to `419ee36c1e310469209f7b83c096ec8aea448386`. Blue and purple boxes denote bounded local capabilities; gray denotes inventory or a non-comparable process state; red denotes a named missing bridge. The P24–P28 numerical statements are batch-specific. No row is farther along a common Route, and no result transfers across rows.

The 1,126 count in figure 4 is deliberately an inventory fact, not a productivity metric. In the current conversion snapshot, 1,089 reader pages are TeX-derived and 37 lack canonical full text; there are no PDF-text fallbacks. These fields make it possible to locate materials and identify conversion limits, but they do not assess proof quality, novelty, or scientific progress. The historical mathematical synthesis remains separately bounded to the source-corpus snapshot recorded in section 3.3.

Phase I was organized as one exploratory entry point and five family-specific sessions, rather than as a prospectively balanced breadth-first candidate portfolio. It consequently does not benchmark breadth against depth, measure time-to-progress, or establish that one search policy is globally optimal. Its defensible portfolio-level lesson is narrower: repeated work around early arithmetic-origin and ownership boundaries did not yield a documented same-object $A_0+A_1+A_2$ chain. That observation motivates, but does not validate, the breadth-first policy in the prospective Layer-2 contract.

The portfolio supports two conclusions. First, Phase I did *not* identify an already formed candidate that can simply be extended to a proof. More precisely, within the source-bound record, no top-level direction is documented as carrying an integrated $A_0+A_1+A_2$ chain in one natural candidate. This is a statement about the record and its named objects, not an impossibility theorem about every system in the corresponding family.

Second, Phase I did not amount to empty failure. It preserved reusable structures: a same-object analytic determinant chain in the Logistic line, periodic classification and geometric interfaces in the Hénon and Symplectic lines, exact finite-system controls in the Symbolic line, and continuous-time ownership tests in the Flow line. The methodological result is the identification of bridges that cannot be filled by prose alone: arithmetic source, clock/repetition convention, analytic owner, and operator owner must be frozen together.

4 From family scan to candidate engineering

The principal Phase-I recommendation is not that breadth is universally superior to depth. The record is neither an optimization benchmark nor a controlled comparison of research strategies. Rather, repeated early boundaries, combined with the non-additivity of evidence across objects, motivate a change in emphasis: explore a broader portfolio of *candidate architectures*, but subject every architecture to low-cost, early, same-object tests before investing deeply in an A0/A1-limited system.

Here, broader does not mean a random sweep over generic dynamical forms. It means a portfolio of lineage-compatible descendants of the prime-symbolic starting point, each with a documented deformation, dimensional lift, geometric realization, or stated preservation failure. A system that lacks that entry condition remains useful as an external positive or negative control, but it does not enter the main candidate portfolio.

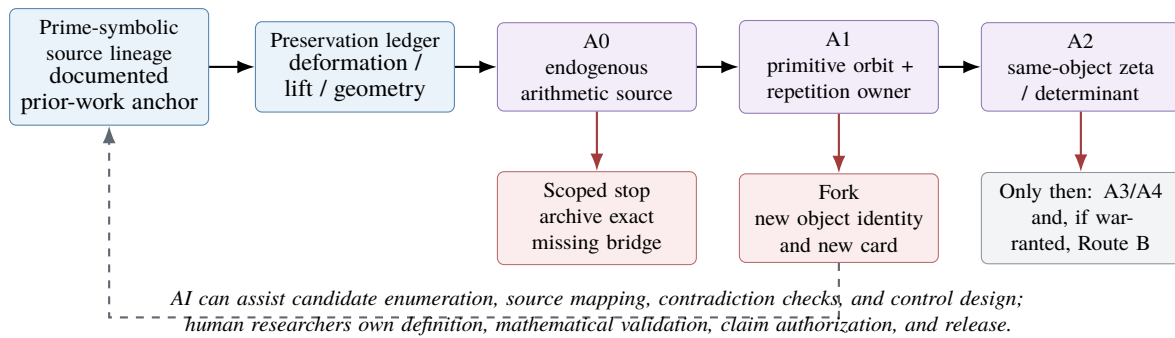


Figure 5: From lineage-constrained family scanning to candidate engineering: proposed human–AI protocol. Every candidate enters through a documented prime-symbolic source lineage and must state its preservation, deformation, lift, or geometric-realization mechanism before A0 is discussed. The flowchart is a methodological recommendation from the Phase-I record, not an experimental proof that breadth dominates depth. A failed bridge is archived at its exact scope; a genuinely different construction is forked rather than inheriting credit.

4.1 A candidate card before target matching

Every prospective candidate should receive a small, versioned candidate card before a Route label is discussed. The card in table 4 asks the following questions.

The card makes a crucial research economy possible. It places cheap, disqualifying ownership tests before expensive operator construction, long numerical searches, or a polished narrative. A Logistic-style analytic chain is then valuable as a control rather than as an invitation to resume closed root hunting; Hénon and Symplectic structures become possible geometric ingredients to test inside a new object, not inherited Route credit; and Symbolic coding is admissible only when it arises from the same candidate’s geometry and clock.

The same card is designed to make a common historical reversal detectable early: selecting a generic high-dimensional or conservative system first, then decorating it with primes. For a non-autonomous proposal, the deformation law itself must be specified; for a dimensional or geometric lift, the card must name a projection, semiconjugacy, invariant, return relation, or explicitly open preservation claim. The words high-dimensional, area-preserving, or symplectic are not by themselves a prime-symbolic mechanism.

4.2 Why the next search should be architecturally broader

The roadmap’s Round-2 structural-synthesis direction gives a concrete shape to the proposal: seek a single candidate in which arithmetic structure, symbolic coding, a geometric carrier, closed-orbit/trace data, and any later quantum realization are derived from one construction. This does *not* mean splicing together a symbolic model, a symplectic map, a physical flow, and an operator with unrelated owners. It means designing a new object so that the relations can be checked rather than narrated.

Table 4: Candidate-card template for a lineage-constrained, *same-object* search. The card records early stop conditions rather than accumulated Route credit.

Field	Question and early stop condition
Object identity	What exact phase space, map or flow, parameter regime, measure, and normalization are under study? Stop if the object changes between sections.
Lineage and preservation	Which prime-symbolic prior-work object is the ancestor? Is this a non-autonomous deformation, conservative dimensional lift, or geometric realization; and which observable, coding, admissibility rule, or return structure survives? Stop before A0 if this ledger is absent; treat the system as an external control.
Arithmetic source	What internal mechanism produces the relevant arithmetic data? Stop if prime/zero data are inserted as an external selector.
Clock and repetition	What fixes a primitive orbit, period/roof/action, and the repetition law? Stop if a symbolic clock substitutes for a distinct physical clock without proof.
Analytic owner	Which transfer operator, zeta/determinant, domain, and normalization belong to this object? Stop if an analytic object is borrowed from a calibrator.
Controls	What nearby positive and negative controls distinguish a mechanism from a fitted coincidence? Stop if no falsifying comparison is available.
Operator owner	Which Hilbert space, operator, domain, and symmetry would support a later quantum claim? Defer, rather than assert, if they are absent.
Decision	Which gate is directly evidenced, open, not-testable, inapplicable, stopped, or forked? Never infer a label from adjacent work.

There is an important distinction between wider search and lower standards. Candidate diversity is useful because Phase I exposed several orthogonal failure modes. The screening standard becomes stricter, not looser: every new architecture must first survive A0, then A1, then A2 with the same identity. An early scoped negative result is a successful search output when it rules out a named construction under named conventions; it can reduce the risk that the next round repeats the same ownership transfer in new vocabulary.

5 Discussion, limitations, and conclusion

The distinctive contribution of this work is methodological rather than a claim of mathematical closure. Its first contribution is the framework in figure 1: the mathematician fixes a prime-symbolic research origin and scientific authority; a jointly drafted, human-approved AGENTS.md contract makes the permissible search, decision labels, breadth-first strategy, and stopping conditions explicit; and AI returns a source-bound handoff rather than extending the project autonomously. The obligation roadmap then treats the HP programme as non-compensatory. A collection of impressive local features does not add up to a candidate. Negative controls, stops, and inapplicability become informative because they identify exactly where a bridge is absent. The versioned repository preserves provenance, source identity, and the distinction between historical research materials and this later synthesis.

5.1 Extensibility beyond the present case study

Extending a fixed mathematical origin across dynamical forms. The framework does not prescribe one privileged map, flow, or dimension. Starting from the prime-symbolic origin in figure 3, a future portfolio may examine a non-autonomous deformation, a dimensional or conservative lift, a symplectic/geometric realization, a suspension flow, or a quantum-dynamical counterpart. The extension is admissible only when its card names what is preserved, transformed, lifted, or shown to fail. This makes system-family diversity a disciplined expansion of a stated mathematical origin, rather than a post-hoc search for a system to which primes can be attached.

Extending to a new human-defined basic idea. The framework is also not permanently tied to prime-symbolic dynamics. A mathematician may initiate a different basic mathematical idea based on judgement, conceptual reframing, intuition, imagination, creative structural insight, or a new observation; Layer 1 then records the new origin, permitted data, and admissible transformations before AI-assisted search begins. The assignment of problem selection, conceptual reframing, and scientific authorization to the mathematician is a governance choice: the framework does not assume a reliable autonomous procedure for recognizing a mathematically fertile new starting idea. It is not an empirical claim that AI systems lack every form of mathematical creativity. Its purpose is to make changes of research origin explicit, reviewable, and attributable rather than implicit in a generated narrative.

Extending through cumulative knowledge infrastructure. The P1 Wiki is designed to retain a candidate's source lineage, positive local results, external and negative controls, scoped failures, stopping conditions, and links back to canonical evidence. Such records make a failed construction useful to a later agent or mathematician without turning it into transferable Route credit. Preserving the relation among inputs, versions, procedures, and outputs follows general reproducibility guidance [9]; the findability and reusability aspiration is aligned with the FAIR principles [12]. The Wiki itself is a navigation and provenance layer, not a proof checker, archival guarantee, or assertion that all included materials satisfy FAIR criteria. Its precise reuse hierarchy and reader path are given in appendix B.

Table 5: Three extensibility modes of the proposed framework. Each mode changes the search space or the reuse relation explicitly; none permits untracked transfer of Route credit or mathematical claims.

Mode	Required human–AI record	Non-transfer boundary
Within-origin dynamical expansion	A candidate card identifies the prime-symbolic ancestor, named preservation/deformation/lift/geometry mechanism, permitted family and dimension, controls, and first gate. AI may enumerate and audit the declared descendants.	No generic map, flow, or operator becomes a main candidate by receiving prime labels afterward.
New-origin research restart	A mathematician records a new basic mathematical idea, scientific question, permitted data, admissible transformations, and a new or revised version-frozen contract. AI may map prior knowledge and execute only the newly authorized search.	The new programme does not silently inherit candidate identity, evidence, or Route credit from the prime-symbolic case study.
Cumulative reuse	The Wiki links a later researcher to the source lineage, positive result, control, failure, stop condition, and canonical evidence package; a new card records the precise reuse. AI may navigate, cross-check links, and prepare comparison ledgers.	A navigation summary, clean build, or past local result is not a proof source, a Route pass, or an authorization to reopen a stopped line.

Several limitations follow. The paper is not an independent proof audit of the component results; its detailed claims remain constrained by their named source records. It is not a benchmark comparing AI systems or research strategies. It does not establish that a broader portfolio is optimal. Rather, the Phase-I record motivates a broader, gate-aware design recommendation in place of unbounded deepening of previously limited objects. Finally, it is not a proof of RH, a complete HP realization, an exact prime-power trace formula, a completed- Ξ determinant identity, or a spectral identification of Riemann zeros.

Within those limits, Phase I supplies a useful research result: it turns failure modes into design constraints. The next step is not to pick a winner from six incomparable lines. It is to construct and screen new *same-object* candidate architectures that begin either from a documented prime-symbolic lineage or from a newly approved Layer-1 mathematical origin, then state an explicit preservation/deformation/lift mechanism, arithmetic origin, clock/repetition ownership, determinant ownership, controls, and a route decision. If a future candidate survives those constraints, its positive claims should be easier to audit. If it fails, the recorded failure may still inform the next search.

Table 6: Reader map for the detailed Phase-I record. Start at the named P1 Wiki direction, then follow its index, conclusions, roadmap, and sources pages to the controlling package. The table is a navigation aid, not a replacement for canonical evidence.

Track	P1 Wiki entry	First question for reuse
Early free entry	directions/zeta_mvp0/	Which local trace/operator structure is claimed, and where is the endogenous prime-time bridge still open?
Session 1: Logistic	directions/logistic_dynamics/	Which return–roof–transfer–Fredholm components share one owner, and which target obligations remain absent?
Session 2: Hénon	directions/henon_dynamics/	Which restricted periodic/geometric result is being reused, with what object-specific limit?
Session 3: Symplectic	directions/symplectic_map/	Is this a scoped geometric interface or an actual Route-typed candidate?
Session 4: Symbolic	directions/symbolic_dynamics/	Does a finite symbolic result own a geometric carrier, physical clock, and analytic/operator object?
Session 5: Flow	directions/flow_systems/	Are clock, closed orbit, repetition, determinant, trace, and operator owned by one continuous-time candidate?

Research materials and reproducibility

The versioned research record, protocols, source materials, roadmap asset, and buildable paper package are available at <https://github.com/maris205/hilbert-polya-structure> [11]. The historical detailed synthesis is tied to the source-corpus snapshot named in section 3.3; before an external submission, the author should create an immutable Git tag or release and cite that tag/commit in the venue-specific version. The repository URL alone should not be treated as a permanent archival identifier.

Ethics, authorship, funding, conflicts, and AI-use disclosure

Ethics. This work did not involve human participants, personal data, or animal research. **Author contributions (CRediT).** Liang Wang: conceptualization, methodology, investigation, visualization, writing—original draft, and writing—review and editing. **Funding.** No funding statement was supplied for this research-record version. **Competing interests.** A competing-interests declaration must be completed by the author before an external submission. **AI-use disclosure.** AI assistance was used for framework and protocol drafting, corpus navigation, source mapping, consistency-oriented drafting, visualization planning, and review-checklist preparation. The human author retains sole responsibility for mathematical interpretation, validation, claim approval, authorship, and release.

A Detailed Phase-I research record

This appendix keeps the direction-level detail that is deliberately compressed in the conference-oriented main text. Each paragraph is a source-bound summary of a named record, not a new theorem or a retroactive common Route evaluation.

A.1 Early free exploration: zeta_mvp0

The early `zeta_mvp0` programme preserves local counting structures, a fixed self-adjoint compact-resolvent operator convention, and a certified local relative-trace construction. These are meaningful local operator

and trace results. Its global claim ledger nevertheless leaves the endogenous prime-time mechanism open: no licensed internal rule supplies the relevant $r \log p$ -type timing with von Mangoldt weights. The corresponding zero-spectral relation and RH remain unauthorized. Thus an internal local label must not be translated into P1 Route-A credit merely because the vocabulary resembles a later route. The line also includes an imported RH archive whose source-preserving status is itself instructive: nearby material does not become native evidence merely by being stored in the same directory [11].

A.2 Session 1: Logistic Dynamics

The Logistic line supplies the clearest preserved example of a *same-object* analytic chain. At a frozen exact algebraic parameter, its record connects first-return structure, an intrinsic polar roof, complex inverse branches, boundary/partition bookkeeping, a nuclear transfer operator, a Fredholm determinant, and growth/order information. This is valuable precisely because the objects and normalizations are retained together. The same record also draws its boundary: the chain does not provide the Riemann-target arithmetic source, target-level global closure, or a natural quantization. Root hunting and Riemann-zero comparison are therefore stopped for that object. The methodological use of the line is as a seed/control for object ownership, not as an unfinished RH proof [11].

A.3 Session 2: Hénon Dynamics

The Hénon record studies restricted families through periodic classification and related low-/higher-dimensional geometric diagnostics. It preserves scoped rational/integer, affine-model, and period-spectrum results. Its Route-A interpretations remain object-specific: arithmetic relevance is weak; primitive-orbit information may be weak or fail for the particular object; and later analytic or lift obligations are absent or not testable in the named scope. Later frozen/admitted contracts were retained without being misrepresented as completed papers. This makes Hénon a potential Round-2 geometric bridge or Poincaré-section ancestor for a new construction, not an inheritance mechanism for a different candidate's analytic or arithmetic credit [11].

A.4 Session 3: Symplectic Map

The Symplectic session produced five scoped interfaces and diagnostic tools around conservative structures, periodic schemes, and qPI-related contexts. Its formal disposition is NOT_APPLICABLE for the route schema used elsewhere. That status is not a scientific failure, and it is not a pass. It means that no fixed candidate has been supplied to own simultaneously an endogenous prime-to-orbit mechanism, a logarithmic clock/repetition law, a target determinant, and a natural operator lift. The session is consequently valuable as a carrier/diagnostic resource, but it cannot be displayed as partial gate closure on a shared HP scorecard [11].

A.5 Session 4: Symbolic Dynamics

The Symbolic session provides exact, finite, autonomous deterministic-system results with a particularly audit-friendly evidence pattern. Its exact-five internal-complete and external-hold labels are process states, not a claim of global analytic continuation or an operator realization. The important structural boundary is that a finite symbolic system, however clean its algebra, does not automatically own a geometric carrier, physical clock, primitive closed-orbit convention, global determinant, or quantum operator. For Round 2, symbolic structure is therefore most promising when it is derived from a candidate's geometry rather than attached to a distinct object as an independent unit-roof model [11].

A.6 Session 5: Flow Systems

The Flow session turns the ownership issue into an explicit continuous-time stress test. Across its P24–P28 forms, it records local trace results, negative controls, and obstructions involving roofs, inverse limits, factorization, and operator ownership. Its bounded final ledger reports positive arithmetic $A2 = 0/5$ and Route-B invocation $= 0/5$ for these five forms. The scope is crucial: this is neither a verdict on all flows nor a seventh top-level Phase-I direction. Under the *same-object* criterion used here, its reusable design lesson is that clock, primitive orbit, repetition law, determinant, trace regime, and operator require common ownership or an explicitly established identification, rather than an unsupported combination of a symbolic clock, physical orbit, borrowed determinant, and separate operator [11].

B Repository organization, evidence packages, and reuse boundaries

The repository is organized so that discovery, evidence, and prospective work are useful without being conflated. The P1 Wiki is an Obsidian-style navigation and cross-link layer; it makes the six directions, their conclusions, related records, and source cards discoverable. It does not supersede canonical TeX or PDF, source data, proof objects, locked inputs, receipts, claim ledgers, or formal route determinations. The current conversion manifest explicitly records this boundary.

```

hilbert-polya-structure/
+-- p1_wiki/
|   +-- README.md; 00-agent-start.md; 01-status-and-claim-vocabulary.md
|   +-- 02-cross-stream-relationships.md
|   +-- directions/<six routes>/
|       |   +-- index.md; conclusions.md; roadmap.md; sources.md
|       |   +-- papers/<route--package>/{index.md,fulltext.md}
|   +-- collections/; meta/; tools/
|   +-- phase1_ai_guided_exploration/
+-- zeta_mvp0/ ... flow_systems/          source project materials
+-- arithmetic_symplectic_flow/          prospective Round-2 workspace

```

The first line of use is intentionally lightweight: read the P1 Wiki README, then 00-agent-start, the claim vocabulary, and the cross-stream relationship page; next enter one direction through its index, conclusions, roadmap, and sources pages. A reader who needs a theorem hypothesis, exact evaluator, frozen input, or canonical manuscript must continue to the linked source package rather than cite the derived Markdown rendering as if it were the proof source. The paper-index and fulltext pages are reader aids with explicit source metadata, not independent validation.

Table 7: Authority hierarchy for reuse. The levels are complementary rather than interchangeable.

Layer	Permitted use	Cannot establish
P1 Wiki navigation	Discover directions, compare scoped conclusions, trace source lineage, preserve cross-links and negative records.	Mathematical validity, a route pass, or replacement of a canonical source.
Source packages and receipts	Inspect exact hypotheses, proof dependencies, data, frozen contracts, evaluators, releases, and local result scope.	Transfer of a result to another object or external peer review.
This conference paper	Communicate the framework, portfolio-level synthesis, figures, and nonclaims in a compact form.	An independent re-proof of every component result or an archival identifier for a mutable repository.
Prospective Round-2 workspace	Maintain a revisable AGENTS.md, plan.md, candidate cards, and bounded new research records.	Retroactive change to Phase-I evidence or authorization to continue without human review.

The prospective workspace is deliberately separated from the historical evidence layer. It can use a standard package schema such as README.md, paper.md, candidate-card.md, claim-ledger.md, and an evidence directory, while remaining mutable until a contract, inputs, results, and release status are frozen. This is why a new experiment cannot silently overwrite an earlier negative result or reclassify an old route disposition.

Two lightweight checks support navigation maintenance:

```
python3 p1_wiki/tools/build_paper_corpus.py --check
python3 p1_wiki/tools/verify_wiki_links.py
```

They validate conversion and local-link integrity only. A passing check does not establish correctness of mathematics, the adequacy of a candidate card, or independent reproducibility. Before external circulation, the author should freeze the cited state in an immutable tag or archival release [11].

C Candidate-card template and non-transfer checks

The compact candidate card in section 4 is the operational template. For the archival research record, its non-transfer checks are:

- C1. Identity.** Freeze phase space, map/flow, parameters, measure, and normalization. A changed object receives a new card.
- C2. Lineage.** Name the prime-symbolic ancestor and the precise deformation, dimensional lift, geometric realization, or preservation failure. Without this ledger, classify the system as an external control before A0.
- C3. Arithmetic and clock.** State permitted data, the endogenous arithmetic mechanism, primitive convention, time/roof/action, and repetition law. Do not replace a physical clock with a distinct symbolic clock.
- C4. Analytic and operator owners.** Name the transfer operator, zeta/determinant, domain, normalization, and any later Hilbert-space/operator pair. Do not borrow these from a proxy.
- C5. Controls and decision.** Record positive and negative controls, direct evidence, and an explicit OPEN, STOP, FORK, or EXTERNAL CONTROL decision. No neighboring Session supplies credit.

D Evidence register and asset provenance

Repository and source boundary. The public research record is <https://github.com/maris205/hilbert-polya-structure>. The detailed Phase-I synthesis is source-bound to commit 419ee36c1e310469209f7b83c096ec8aea448386; the new conference-oriented package is a later, separately versioned presentation layer. The repository’s P1 Wiki provides the six-direction index, claim vocabulary, cross-stream non-transfer rules, source cards, and links to the original project materials [11].

Prior-work lineage guide. The prime-symbolic search origin is documented in `flow_systems/docs/prior_work/README.md` and mirrored for the Round-2 workspace at `arithmetic_symplectic_flow/docs/prior_work/README.md`. These guides are used here as internal provenance for the constrained search genealogy, not as proof that any Logistic, Hénon, or symplectic candidate already realizes prime dynamics.

Research-framework asset. Figure 1 uses `assets/research_framework.png`, a byte-identical release-local mirror of the author-supplied P1 Wiki source `p1_wiki/research_framework.png`. Its SHA-256 is

```
da9e89f253991fb02db9ae11c2b13a51
2a4a404f720bf9abe03ff3853887f4a8
```

The diagram represents an author-proposed prospective governance framework. It is not evidence for a mathematical claim, a Route verdict, or a statement that Phase-I projects historically shared one workflow contract.

Roadmap asset. Figure 2 uses `assets/rh_roadmap.png`, a byte-identical mirror of the tracked Round-2 roadmap reference. Its SHA-256 is

```
f5e70c5120474702e4f0715bce90c7a
88fd5936164594e8b37a2ee0993549d1d
```

The asset is identified as author-proposed and is not substituted silently for the older Phase-I roadmap image used by the historical internal working paper.

External sources. The external bibliography supports only the background and methodology claims made in the main text: AI-assisted conjecturing, evaluator-coupled program search, bounded neuro-symbolic geometry, Lean-grounded proof search, Hilbert–Pólya/trace-formula context, dynamical zeta context, and the current official RH status. Detailed local mathematical statements are sourced to the repository record rather than inferred from the external references.

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